Signal Processing

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March 25,2026

Question

Let a signal $a_1 \sin(\omega_1 t + \phi_1)$ be applied to a stable linear time-invariant system. Let the corresponding steady state output be represented as $a_2 F(\omega_1 t + \phi_2)$. Then which of the following statements is true? (EE 2007)

(a) F is not necessarily a "sine" or "cosine" function but must be periodic with $\omega_1 = \omega_2$

(b) F must be a "sine" or "cosine" function with $a_1 = a_2$

(c) F must be a "sine" function with $\omega_1 = \omega_2$ and $\phi_1 = \phi_2$

(d) F must be a "sine" or "cosine" function with $\omega_1 = \omega_2$

Theoretical Solution

In the given question the input x is a singular scalar value. However, for the sake of developing a rigorous generalized solution the following analysis considers x to be an N dimensional complex sinusoidal vector.
For arriving at the final output, since the given input x can be treated as the imaginary part of one dimensional complex sinusoid vector , the same operation is applied on the output $\mathbf{y}$, taking imaginary part to get the final answer.

Theoretical Solution

Every LTI system is uniquely defined by its impulse response vector, $\mathbf{h} = [h[0], h[1], \ldots, h[N-1]]^T$.
For any periodic input signal $\mathbf{x}$, the output $\mathbf{y}$ is algebraically defined by the circular convolution sum. The n-th sample of the output is:

$$y[n] = \sum_{m=0}^{N-1} h[m]x[(n-m) \pmod N] \tag{1}$$

This equation simply states that the output is a weighted sum of shifted versions of the input.

Theoretical Solution

Instead of writing a summation for every single sample $y[0], y[1], \ldots, y[N-1]$, we can write the entire system as a single matrix equation: $\mathbf{y} = \mathbf{C}\mathbf{x}$.
By expanding the convolution sum, the system matrix $\mathbf{C}$ is populated by cyclic shifts of the impulse response $\mathbf{h}$:

$$\begin{bmatrix} y[0] \\ y[1] \\ \vdots \\ y[N-1] \end{bmatrix} = \begin{bmatrix} h[0] & h[N-1] & \ldots & h[1] \\ h[1] & h[0] & \ldots & h[2] \\ \vdots & \vdots & \ddots & \vdots \\ h[N-1] & h[N-2] & \ldots & h[0] \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix} \tag{2}$$

Because the system is **Time-Invariant**, every row is just the row above it shifted to the right. This specific structure is called a **Circulant Matrix**.

Theoretical Solution

Now, let our input vector $\mathbf{x}$ be a complex sinusoid with frequency ω_1 (represented by discrete index k). Its m-th element is $x[m] = e^{j\frac{2\pi}{N}km}$. We compute the matrix multiplication $\mathbf{y} = \mathbf{C}\mathbf{x}$. Looking at the n-th row:

$$y[n] = \sum_{m=0}^{N-1} h[m] e^{j\frac{2\pi}{N}k(n-m)} \tag{3}$$

Using exponent rules ($e^{A-B} = e^A e^{-B}$), we can factor the term $e^{j\frac{2\pi}{N}kn}$ right out of the sum, because it doesn't depend on m:

$$y[n] = \underbrace{e^{j\frac{2\pi}{N}kn}}_{\text{Input } x[n]} \cdot \underbrace{\left(\sum_{m=0}^{N-1} h[m] e^{-j\frac{2\pi}{N}km} \right)}_{\text{Complex Constant } \lambda} \tag{4}$$

Theoretical Solution

This simplifies the entire matrix operation into a simpler form:

$$\mathbf{y} = \lambda \mathbf{x} \tag{5}$$

Because multiplying the matrix $\mathbf{C}$ by the vector $\mathbf{x}$ just returns the vector $\mathbf{x}$ scaled by a number λ, the sinusoid $\mathbf{x}$ is an **eigenvector** of the system.

Theoretical Solution

The constant λ is a complex number representing the system's Frequency Response. We can write it in polar form as $\lambda = |H|e^{j\theta}$.
Substituting this into our eigenvector equation gives us the exact relationship between the input and output physical signals:

$$\text{Output} = \left(|H|e^{j\theta}\right) \cdot a_1 \sin(\omega_1 t + \phi_1) = (a_1|H|)\sin(\omega_1 t + \phi_1 + \theta) \quad (6)$$

By comparing this result to the proposed output $a_2 F(\omega_2 t + \phi_2)$, we conclude:

1. **The Function (F):** The output shape remains a sine wave.
2. **The Frequency (ω):** The variable ω_1 is locked inside the eigenvector $\mathbf{x}$. Because the equation is $\mathbf{y} = \lambda\mathbf{x}$, the vector itself cannot change. Therefore, ω_1 **must equal** ω_2.
3. **Amplitude and Phase:** The eigenvalue dictates the new amplitude ($a_2 = a_1|H|$) and new phase ($\phi_2 = \phi_1 + \theta$).

Therefore, statement **(d)** is mathematically proven to be true.

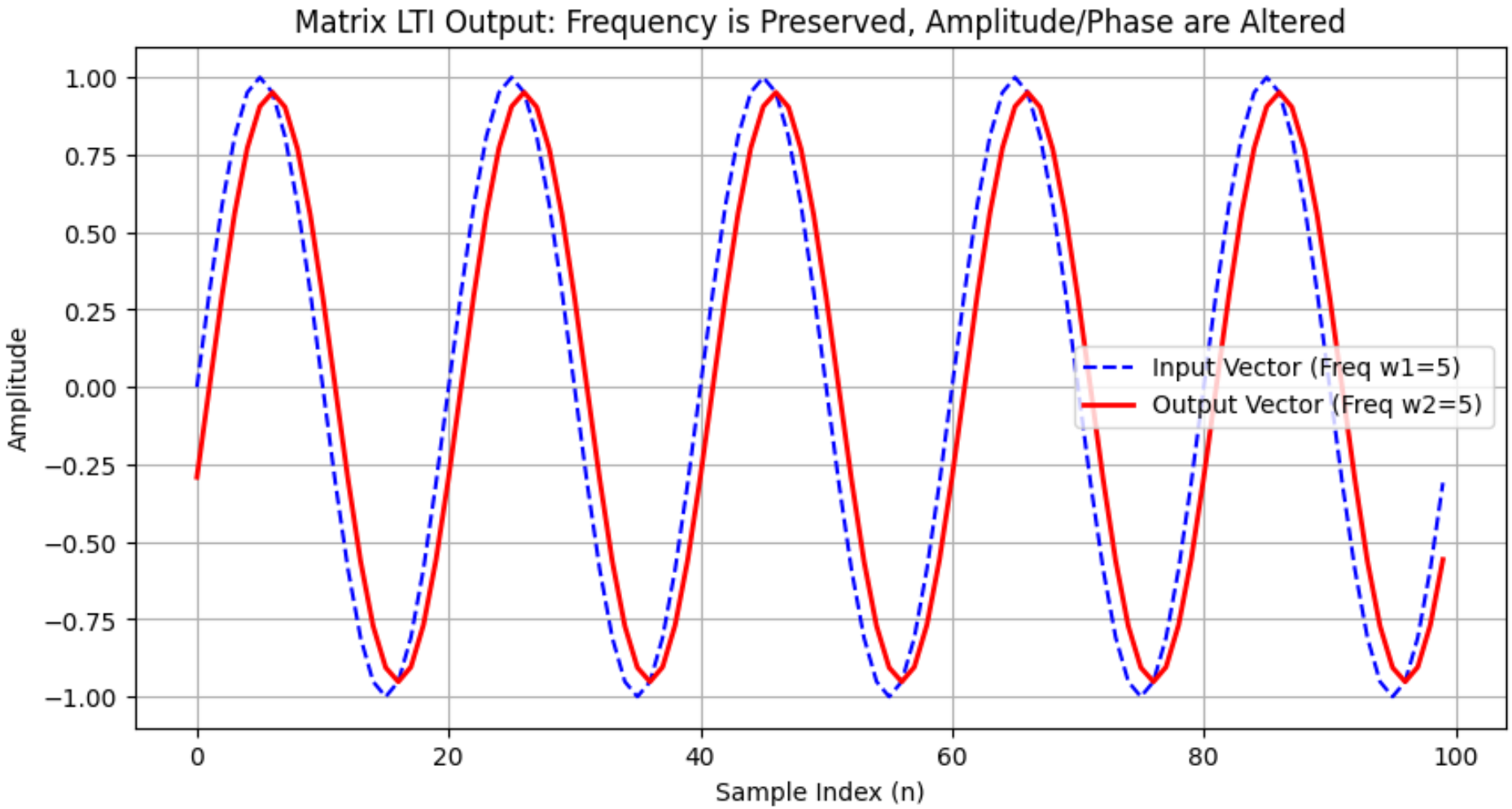
Python Code

The following script builds the algebraic impulse response, converts it to a Circulant Matrix, and verifies the eigenvector property via matrix multiplication.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.linalg import circulant

N = 100
n = np.arange(N)
k = 5 # Frequency: 5 cycles per window
x = np.sin(2 * np.pi * k * n / N)
h = np.zeros(N)
h[0:4] = [0.4, 0.3, 0.2, 0.1] # Simple causal filter
C = circulant(h)
y = C @ x
```

Python Code

```
plt.figure(figsize=(10, 5))
plt.plot(n, x, label=f'Input Vector (Freq w1={k})', linestyle='--
    ', color='blue')
plt.plot(n, y, label=f'Output Vector (Freq w2={k})', linewidth=2,
     color='red')
plt.title('Matrix LTI Output: Frequency is Preserved, Amplitude/
    Phase are Altered')
plt.xlabel('Sample Index (n)')
plt.ylabel('Amplitude')
plt.legend()
plt.grid(True)
plt.show()
```

Plot-Using only Python